

SUMMARY

This video explains an advanced synthesis of Minimal Deterministic Finite Automata (DFA) construction strategies within Theory of Computation. The focus includes substring detection, parity conditions, compound prefix-length constraints, and refined minimization logic.

Key Takeaways:

The $m + n + 1$ rule computes states for prefix-plus-modulo constraints.

Languages equating counts of “01” and “10” are regular because their difference is bounded.

Many textbook DFA diagrams are not minimal.

True minimization requires structural reasoning, not memorization.

I. SUBSTRING CONSTRAINTS: DEBIT AND TRIBIT

Substring DFAs are constructed by building the shortest acceptance path and preserving partial progress upon mismatches.

I.1 Debit Substrings (“00” or “11”)

Objective: Accept strings containing either “00” or “11”.

States Required: 4

Logic:

Track first occurrence of 0 or 1.

If two identical consecutive symbols appear, move to final state.

Final state self-loops on both inputs (0,1).

I.2 Tribit Substrings (“000” or “111”)

States Required: 6

Logic:

Track first, second, and third consecutive identical symbols.

If interrupted (e.g., “00” followed by “1”), transition to state tracking new character rather than resetting.

Final state self-loops on all inputs.

II. EQUALITY OF SUBSTRING OCCURRENCES

Language Definition:

Number of occurrences of “01” equals number of occurrences of “10”.

Recognizability Insight:

For any binary string, the difference between counts of “01” and “10” is always -1 , 0 , or 1 . Since this difference is finite, the language is regular.

Language Composition:

Accepts:

ϵ
0
00
1
11
010
101

Rejects:

01
10

Minimal DFA:

States Required: 5

Structure:

Initial state is final (accepts ϵ).

Separate branches handle strings starting with 0 and 1.

Transitions maintain balance between substring counts.

III. PARITY AND MODULO CONSTRAINTS

Languages based on parity of substring occurrences use cyclic constructions.

Language Constraint	States	Logic
“10” occurs Even times	4	Accepts ϵ and zero occurrences
“10” occurs Odd times	4	Complement of Even DFA

Odd DFA is obtained by swapping final and non-final states of Even DFA.

IV. COMPOUND CONSTRAINTS: START AND LENGTH DIVISIBILITY

Language Structure:

Strings must start with substring S of length m.

Total length $\equiv r \pmod n$.

IV.1 The $m + n + 1$ Rule

Total States = $m + n + 1$

Where:

m = prefix length

n = modulus

1 = dead state

IV.2 Application Examples

Starting Substring	Length Constraint	Calculation	Total States
"a" (m = 1)	Divisible by 3 (n = 3)	$1 + 3 + 1$	5
"ba" (m = 2)	$2 \pmod 4$ (n = 4)	$2 + 4 + 1$	7
"aba" (m = 3)	$2 \pmod 5$ (n = 5)	$3 + 5 + 1$	9

Dead state handles prefix violations.

Remaining states form modulo-length cycle.

V. POSITIONAL "FOLLOWED BY" CONSTRAINTS

V.1 Every 'A' is Followed by 'B'

Minimal States: 3

States:

1. Final state (valid sequences and ϵ)
2. Non-final state after reading 'A'
3. Dead state for violation

Logic:

After 'A', next symbol must be 'B'.
If 'A' followed by 'A', transition to dead state.

V.2 Every 'A' is Never Followed by 'B'

States Required: 3

If substring "AB" appears → dead state.

V.3 Every 'A' is Followed by "BB"

States Required: 4

Additional state required to track first 'B' after 'A'.

VI. TECHNICAL CRITIQUE: GATE 2004 AMBIGUITY

Problem Context:

DFA intended for language:

Every 'A' is followed by at least two 'B's.

Issue:

If 'A' is absent, any number of 'B's should be accepted.
Exam diagram did not clearly mark all necessary final states.

Conclusion:

Diagram was incomplete.
Ambiguity likely led to universal marking.

VII. STRATEGIC MINIMIZATION PRINCIPLES

Build shortest acceptance path first.
Preserve partial progress on mismatch.
Use complement logic for Even/Odd problems.
Apply $m + n + 1$ rule for prefix-plus-modulo languages.

VIII. PRACTICE PROBLEMS

1. GATE 2017
Minimum states for $L = \{ w_1 a w_2 \mid |w_1| = 2, |w_2| \geq 3 \}$.
2. GATE 2011
Minimum states to recognize a^n where n is multiple of k ($k > 0$).
3. Construct DFA for strings that start with "AB" and end with "BA."

